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Review

Dopamine spillover after quantal release: Rethinking dopamine transmission in the nigrostriatal pathway

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ABSTRACT

The predominance of dopamine (DA) receptors at extrasynaptic vs. synaptic sites implies that DA signaling is by diffusion-based volume transmission. In this review, we compare characteristics that regulate extracellular DA behavior in substantia nigra pars compacta (SNc) and striatum, including regional differences in structure (a 40% greater extracellular volume fraction in SNc vs. striatum) and in dynamic DA uptake (a 200-fold greater DA uptake rate in striatum vs. SNc). Furthermore, we test the assumption of diffusion-based volume transmission for SNc and striatum by modeling dynamic DA behavior after quantal release using region-specific parameters for diffusion and uptake at 37 °C. Our model shows that DA uptake does not affect peak DA concentration within 1 μm of a release site in either SNc or striatum because of the slow kinetics of DATs vs. diffusion. Rather, diffusion and dilution are the dominant factors governing DA concentration after quantal release. In SNc, limited DAT efficacy is reflected in a lack of influence of uptake on either amplitude or time course of DA transients after quantal release up to 10 μm from a release site. In striatum, the lack of effect of the DAT within 1 μm of a release site means that perisynaptic DATs do not “gate” synaptic spillover. This contrasts with the conventional view of DA synapses, in which DATs efficiently recycle DA by re-uptake into the releasing axon terminal. However, the model also shows that a primary effect of striatal uptake is to curtail DA lifetime after release. In both SNc and striatum, effective DA radius after quantal release is ~2 μm for activation of low-affinity DA receptors and 7–8 μm for high-affinity receptors; the corresponding spheres of influence would encompass tens to thousands of synapses. Thus, the primary mode of intercellular communication by DA, regardless of region, is volume transmission.

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Abbreviations: DA, dopamine; DAT, dopamine transporter; [DA]_o, extracellular DA concentration; SNc, substantia nigra pars compacta

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1. Introduction

The nigrostriatal dopamine (DA) pathway extends from the substantia nigra pars compacta (SNc) to the dorsal striatum via the median forebrain bundle. Somatodendritic release of DA in the SNc and as axonal DA release in the striatum are both necessary for basal ganglia-mediated motor behaviors (Robertson and Robertson, 1989; Timmerman and Abercrombie, 1996; Bergquist et al., 2003). Transporters for DA (DATs) are expressed exclusively by DA neurons and are found extrasynaptically on DA axons in striatum and on DA somata and dendrites in midbrain (Ciliax et al., 1995; Nirenberg et al., 1996; Hersch et al., 1997). Moreover, DA receptors are also predominantly extrasynaptic (Sesack et al., 1994; Yung et al., 1995; Hersch et al., 1995; Khan et al., 1998). This implies that intercellular communication by DA requires diffusion-based volume transmission (Fuxe and Agnati, 1991), which is most simply defined as “a functionally significant association of release and receptor sites via extrasynaptic diffusion” (Rice, 2000).

One would expect, therefore, that extracellular DA concentration ($[DA]_o$) and lifetime after release would be regulated by diffusion, as well as uptake. This is indeed the case, although the relative contribution of each process differs between striatum and SNc. In striatum, $[DA]_o$ is often considered to be “uptake limited”, with strong local regulation by the DAT (Stamford et al., 1988; Giros et al., 1996; Floresco et al., 2003), whereas in SNc, the influence of uptake is much less (Cragg et al., 1997), so that diffusion-based volume transmission is expected to dominate (Rice, 2000; Cragg et al., 2001). Recently, however, both of these basic expectations have been challenged. Evidence from physiological recording in midbrain DA neurons suggests limited diffusional contributions to DA transmission in the SNc (Beckstead et al., 2004, 2007), whereas modeling of quantal DA release from DA axons in striatum indicates minimal regulation of synaptic spillover by the DAT (Cragg and Rice, 2004).

In this review, we extend our previously published model of quantal DA release in striatum (Cragg and Rice, 2004) to include somatodendritic release in SNc. Other models that have been used to estimate the sphere of influence of released DA have been limited by the single distance examined (Gonon, 1997; Cragg et al., 2001) or by the evaluation of a population response in which DA diffusion, and thus diffusion distance, is irrelevant (Garris et al., 1994; Sulzer and Pothos, 2000). The

model used here is based on experimentally determined parameters for diffusion and uptake in SNc and striatum and permits comparison of the duration and sphere of influence of $[DA]_o$ after single-vesicle release in both regions. Our findings confirm the comparatively limited role of uptake in regulating DA behavior in SNc vs. striatum and call for a revision of the current concept of synaptic DA release regulation, in which the DAT “gates” DA overflow by re-uptake into the releasing presynaptic terminal.

2. The diffusion model and parameters for quantal DA release

Theoretical concentration–time profiles for quantal DA release at 37 °C were generated using the expression for diffusion from an instantaneous point source (Nicholson, 1985; Cragg et al., 2001) using the program VOLTORO by Dr. C. Nicholson (NYU School of Medicine, USA):

$$C(r, t) = \frac{UC_f}{\alpha(4D^*t\pi)^{3/2}} \exp\left(\frac{-r^2}{4D^*t}\right) \exp(-k't) \quad (1)$$

Quantal DA release was assumed to be instantaneous from a vesicle of volume U with a filling concentration of C_f , such that quantal size, $Q=UC_f$. Diffusion distance, r , was varied from 1–20 μm .

The structural parameters that govern diffusion are the extracellular volume fraction (α) and the tortuosity (λ) of extracellular diffusion paths (Nicholson and Syková, 1998). Experimentally derived values for α , λ , k' , C_f and U for SNc and striatum are given in Table 1. Macroscopic diffusion parameters have been shown to be appropriate to describe microscopic structure, including that surrounding a synapse (Rusakov and Kullmann, 1998). These parameters are assumed to be temperature independent. On the other hand, the diffusion coefficient, D , for DA is temperature dependent; the value used for D at 37 °C was $7.63 \times 10^{-6} \text{ cm}^2 \text{ s}^{-1}$ (Table 1), calculated from D at 32 °C (Nicholson, 1995), assuming a 2% increase per °C.

In experimental studies of DA diffusion and uptake in the SNc, we found that the linear uptake term k' for specific DA uptake was sufficient to fit DA diffusion curves in this region (Cragg et al., 2001). To model nigral DA behavior at 37 °C,

Table 1 – Experimental data used to model DA behavior in striatum and SNc

Parameter	Value	Reference
Striatum		
α	0.21	Rice and Nicholson, 1991
λ	1.54	Rice and Nicholson, 1991
k' for DA	20 s^{-1}	$V_{\max}/K_m = (4.1 \text{ } \mu\text{M s}^{-1})/(0.21 \text{ } \mu\text{M})$ (Ross, 1991)
k' for non-specific uptake	0.007 s^{-1}	Derived from Cragg et al., 2001
Q	9800 molecules	Derived from Jaffe et al., 1998; Cragg et al., 2001
U	$6.5 \times 10^{-20} \text{ L}$	Nirenberg et al., 1997
C_f	250 mM	Derived from Jaffe et al., 1998; Nirenberg et al., 1997; Cragg et al., 2001
D (37 °C)	$7.63 \times 10^{-6} \text{ cm}^2 \text{ s}^{-1}$	Derived from Nicholson, 1995
SNc		
α	0.30	Cragg et al., 2001
λ	1.58	Cragg et al., 2001
k' for DA	0.12 s^{-1}	Derived from Cragg et al., 2001
k' for non-specific uptake	0.007 s^{-1}	Derived from Cragg et al., 2001
Q	14,000 molecules	Jaffe et al., 1998
U	$6.5 \times 10^{-20} \text{ L}$	Nirenberg et al., 1997
C_f	350 mM	Derived from Jaffe et al., 1998; Nirenberg et al., 1997
D (37 °C)	$7.63 \times 10^{-6} \text{ cm}^2 \text{ s}^{-1}$	Derived from Nicholson, 1995

α , extracellular volume fraction; λ , tortuosity factor; k' , linear uptake term; Q, number of DA molecules per vesicle; U, vesicle volume; C_f , intravesicular DA concentration; D, DA diffusion coefficient.

we increased the k' value for SNc in accordance with the temperature-dependence of DA uptake evaluated in striatum (Cragg and Rice, 2004). Similarly, the k' for non-specific DA uptake, determined from measurements of the diffusion of tetramethylammonium (TMA^+) in midbrain (Cragg et al., 2001), was temperature-corrected to 37 °C (Table 1), and was used for “diffusion only” curves generated for both SNc and striatum.

Linear uptake was also used to model DA behavior in striatum (Cragg and Rice, 2004). The relationship between Michaelis-Menten uptake rate and concentration becomes linear when $[\text{DA}]_o$ is substantially below K_m , i.e. $C \ll 0.21 \text{ } \mu\text{M}$ (Ross, 1991). Then, the uptake rate, $V = V_{\max}C/(C + K_m)$, simplifies to first order kinetics with respect to C, such that $V = V_{\max}C/K_m = k'C$, where $k' = V_{\max}/K_m$. Use of linear uptake for striatum in this model is justified because $[\text{DA}]_o$ remains in the nanomolar range following release of a single vesicle, except at distances very near to the site of release (Cragg and Rice, 2004). At those sites, the use of linear uptake overestimates uptake when $[\text{DA}]_o \geq K_m$, because uptake rate increases linearly with increasing concentration ($V = k'C$), rather than approaching an asymptotic value ($V = V_{\max}$). However, this overestimation only strengthens the case for the limited DAT-dependent regulation of peak $[\text{DA}]_o$ after quantal release described below.

A further assumption of our model is that DA vesicle fusion at a DA synapse (or varicosity; Descarries et al., 1996) in striatum or on a DA cell body or dendrite in SNc acts as an instantaneous point source (Cragg and Rice, 2004). The validity of this assumption for SNc is a matter of debate, given the uncertainty of whether release is mediated exclusively by conventional vesicle fusion (Cragg et al., 1997; Chen and Rice, 2001; Falkenburger et al., 2001). Nonetheless, quantal release from DA cells in the SNc has been recorded in acute midbrain slices (Jaffe et al., 1998). We used the reported quantal size (Q)

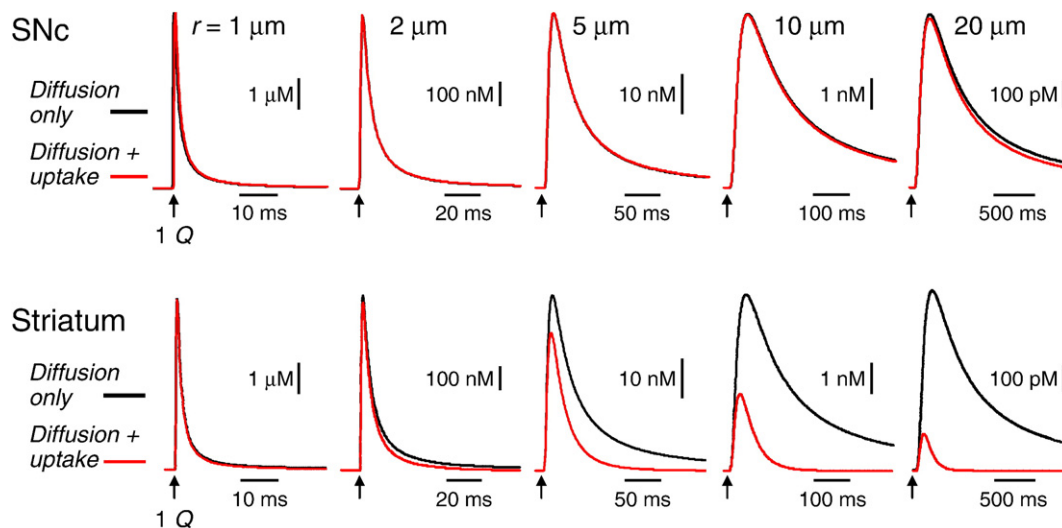


Fig. 1 – DA diffusion following quantal release in SNc and striatum: distance and uptake. Theoretical curves for DA diffusion at 37 °C were generated using Eq. (1) with region-specific parameters for α , λ , k' and normalized Q (Table 1). Curves show dynamic changes in $[\text{DA}]_o$ at diffusion distances, r , of 1–20 μm in SNc (upper panel) and striatum (lower panel), following release of a single DA vesicle (arrows). The effect of DA uptake is indicated by diffusion curves generated with only non-specific uptake ($k' = 0.007 \text{ s}^{-1}$; diffusion only, black lines) vs. region-specific uptake (Table 1; diffusion + uptake, red lines). Scale bars indicate variations in concentration and time across distance. Striatal data are modified from Cragg and Rice, 2004.

of 14,000 DA molecules for simulations of DA behavior in SNc (Table 1), unless otherwise stated.

Because the extracellular volume fraction of midbrain is 40% larger than that in striatum (Rice and Nicholson, 1991; Cragg et al., 2001), the apparent source strength is 40% lower in SNc than in striatum. For initial comparisons of extracellular DA behavior in SNc and striatum, we normalized source strength between regions by a proportional decrease in striatal Q to 9800 molecules (Table 1). The resultant values of C_f (350 mM, SNc; 250 mM, striatum) are within a range consistent with those determined experimentally for DA in cultured neurons (Pothos et al., 1998).

3. DA behavior after quantal release in SNc and striatum

In the SNc, relatively low DAT activity compared to that in striatum leads to the surprising result that peak $[DA]_o$ is minimally affected by uptake even 20 μm from the site of release (Cragg et al., 2001) (Fig. 1, upper panel). By contrast, in striatum, where k' is nearly 200-fold greater (Table 1), an effect of uptake on peak $[DA]_o$ is seen by $r=2\text{ }\mu\text{m}$ (Fig. 1, lower panel). However, whether SNc- or striatum-specific DA uptake is included or not, the dominant influence on $[DA]_o$ is the distance from a release site (note the order-of-magnitude decrease in concentration at each successive r in Fig. 1), as diffusing DA is diluted into an increasing volume defined by $4/3\pi r^3$ (Cragg and Rice, 2004). Overall, therefore, diffusion and dilution are the main factors that govern absolute $[DA]_o$ after quantal release in both SNc and striatum.

3.1. Do perisynaptic DATs in striatum modulate synaptic concentration and “gate” DA spillover?

A striking finding from our simulations of DA behavior in striatum is the complete lack of uptake on peak $[DA]_o$ within 1 μm of a release site (Cragg and Rice, 2004) (Fig. 1, lower panel). Given that the cross-synaptic diameter of a striatal DA synapse is 0.3–0.6 μm (Pickel et al., 1981; Groves et al., 1994; Descarries et al., 1996), this means that DA must necessarily spill over from a synapse or other release site, with its maximum concentration unmodulated by uptake. We tested the hypothesis that this occurs because the rate of diffusion exceeds that of DA uptake near a release site ($r < 5\text{ }\mu\text{m}$; see Fig. 1) by “slowing” DA diffusion through doubling the tortuosity factor, λ , which decreases the apparent diffusion coefficient, D^* , by a factor of four ($D^* = D/\lambda^2$) (Cragg and Rice, 2004). The theoretical slowing of DA diffusion allows more time for DAT-mediated uptake and leads to a slight decrease in peak $[DA]_o$, even 1 μm from the release site. This result confirms that DA uptake normally loses the kinetic competition against diffusion in mediating DA clearance within a synapse. Thus, perisynaptic DATs neither modulate maximum synaptic DA concentration nor gate initial DA spillover (Cragg and Rice, 2004).

3.2. How far does DA diffuse?

The primary role of diffusion in governing $[DA]_o$ at all distances from a release site, whether in SNc or striatum, leads naturally to

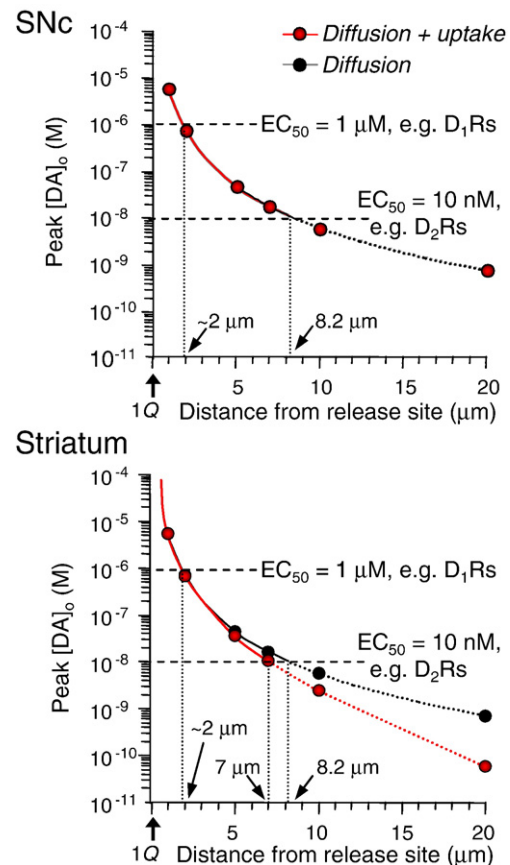


Fig. 2 – Effective radius for DA after quantal release in SNc and striatum. (A) Peak $[DA]_o$ vs. diffusion distance, r , from Fig. 1. The maximum radius at which peak $[DA]_o$ reaches 10 nM (e.g., the concentration required for activation of high-affinity D_2 DA receptors, D_2Rs) was 8.2 μm in SNc or striatum for diffusion only (non-specific k') and in SNc with region-specific uptake with normalized Q to compensate for the larger α of SNc (Table 1). In striatum, region-specific uptake decreases the effective radius to 7.0 μm , resulting in a 40% smaller sphere of influence than for diffusion only. The maximum radius at which peak $[DA]_o$ reaches 1 μM (e.g., for activation of low-affinity D_1 DA receptors, D_1Rs) is $\sim 2\text{ }\mu\text{m}$ in both SNc and striatum. Given the dominance of diffusion over the DAT at this distance from a release site, the sphere of influence for low-affinity DA receptors is unaltered by DAT-mediated uptake. Striatal data are modified from Cragg and Rice, 2004.

the question of how far DA can diffuse. In the absence of uptake (or metabolism) and with sufficient time, DA molecules can diffuse from a release site to the physical boundaries of the brain, i.e., a ventricle or the subarachnoid space. However, the better question is, how far from a release site will DA have physiological consequences? The answer obviously depends on the concentration-sensitivity of DA receptors within a potential sphere of influence. Specifically, the EC_{50} values for activation of D_1 -like and D_2 -like receptors *in vitro* are on the order of 1 μM for low-affinity states and 10 nM for high-affinity states (Richfield et al., 1989; Neve and Neve, 1997). In striatum, the majority of D_1 -like receptors have been suggested to be in a low-affinity state,

whereas D_2 -like receptors are predominantly high affinity (Richfield et al., 1989). Assuming these relative affinities also hold true for SNc, the maximum “effective radius” can be defined as the distance within which release of one quantum of DA reaches high-affinity D_2 receptors at ≥ 10 nM above baseline (Cragg and Rice, 2004).

With this constraint and using the normalized Q values in Table 1, the maximum effective radius for spillover in SNc and in striatum is 8.2 μm , in the absence of uptake (diffusion only) (Fig. 2). Unsurprisingly, given limited DA uptake in SNc, including DAT-mediated uptake in the simulation has no effect on maximum effective radius in SNc (Fig. 2, upper panel). However, in striatum, uptake decreases the maximum effective radius to 7.0 μm (Cragg and Rice, 2004). Although the absolute difference in r seems negligible, the r^3 -dependence of spherical volume means that the sphere of influence of DA, and therefore

the number of receptors activated, is 40% smaller with normal striatal uptake than under conditions of DAT inhibition.

The effective radius for DA acting at low-affinity, predominantly D_1 -like receptors is much smaller than for high-affinity receptors. For receptors requiring 1 μM DA for activation, the maximum effective radius is <2 μm in SNc or striatum (Fig. 2). At this distance from the release site, there is little influence of the DAT on peak $[\text{DA}]_o$ and consequently little influence on the sphere of influence in either SNc or striatum (Fig. 2). Overall, these data indicate that DAT inhibition would have a greater influence on the activation of high-affinity DA receptors than on low-affinity receptors.

3.3. Influence of the DAT on the active lifetime of DA transients

Thus far, we have considered the influence of the DAT on peak $[\text{DA}]_o$ and effective radius of DA transients. A third parameter of importance in DA transmission is active lifetime, i.e., the time DA is available at sufficient concentration to activate DA receptors. The maximum active lifetime is the time during which $[\text{DA}]_o$ remains above the EC_{50} for high-affinity DA receptors, 10 nM, at a given site after quantal release (Cragg and Rice, 2004). This is measured as the difference between the time at which $[\text{DA}]_o$ first reaches 10 nM (onset) and when it falls below 10 nM (offset) (Fig. 3, inset). In SNc, active lifetime is longest at $r \leq 1$ μm , with a maximum lifetime of 100 ms for a single Q , then decreases progressively as r approaches the limit of the effective radius (Fig. 3, upper panel). This pattern is the same whether SNc-specific DA uptake is present or not. In striatum, a similar lifetime pattern is seen in the absence of uptake (Fig. 3, lower panel). However, under conditions of normal striatal DAT activity, uptake curtails the duration of striatal $[\text{DA}]_o$ transients within the effective radius by 50%. Thus, a major role of the DAT in striatum, but not in SNc, is to limit DA lifetime after release. Through this mechanism, the DAT can readily influence DA transmission.

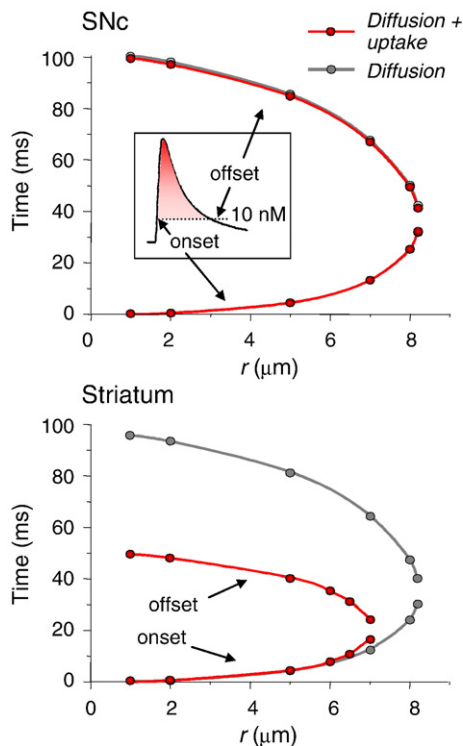
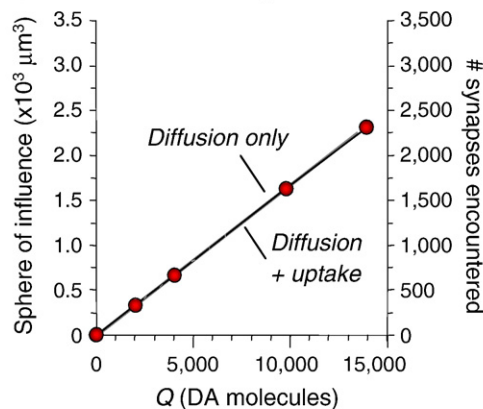
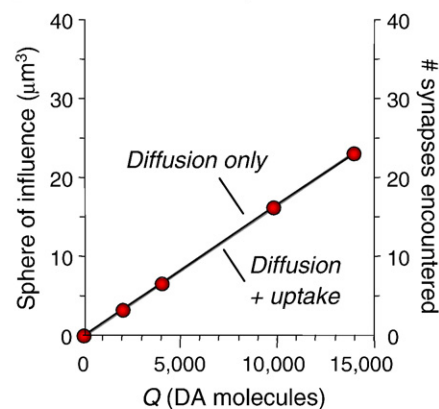


Fig. 3 – Effect of the DAT on active DA lifetime in SNc and striatum. Onset and offset times at varying r after quantal release in (A) SNc and (B) striatum, defined as the times at which a change in $[\text{DA}]_o$ reaches (onset) and then falls below (offset) 10 nM. The inset in (A) indicates these time points for a theoretical DA diffusion curve at $r = 5$ μm in SNc. The data were calculated using the normalized values for Q in Table 1. The effect of uptake on onset and offset times in (A) SNc and (B) striatum was also assessed; compare diffusion only (gray lines) with diffusion + region-specific DA uptake (red lines). In SNc, DAT-mediated uptake does not alter active lifetime. In striatum, however, uptake curtails offset time, leading to a ~50% decrease in active lifetime at all distances within the effective radius. Although the maximum effective radius in SNc is unaltered by the DAT, region-specific uptake in striatum limits the effective radius for activation of high-affinity DA receptors (as also seen in Fig. 2). Striatal data are modified from Cragg and Rice, 2004.

4. Quantal size and the sphere of influence in SNc and striatum

The sphere of influence of DA after quantal release can be calculated from the maximum effective radius for a given set of conditions. We determined the sphere of influence for activation of high-affinity (e.g., D_2 receptors) and low-affinity DA receptors (e.g., D_1 receptors), based on the effective radii for EC_{50} values of 10 nM and 1 μM , respectively, for a range of Q in SNc and striatum (Fig. 4). The range of Q examined was based on available literature indicating that the number of DA molecules per vesicle can vary from 1000 to 14,000 (Garris et al., 1994; Pothos et al., 1998; Jaffe et al., 1998), reflecting the plasticity of the volume and quantal content of catecholamine vesicles (Pothos et al., 1998; Colliver et al., 2000), as well as methodological differences in assessing Q . We did not normalize Q between SNc and striatum for this comparison, but rather used identical Q values for both regions to gain insight into the role of α , as well as the DAT, in the regulation of $[\text{DA}]_o$ after quantal release.

SNc

A ≥ 10 nM, e.g. D_2 RsB ≥ 1 μM, e.g. D_1 Rs

Striatum

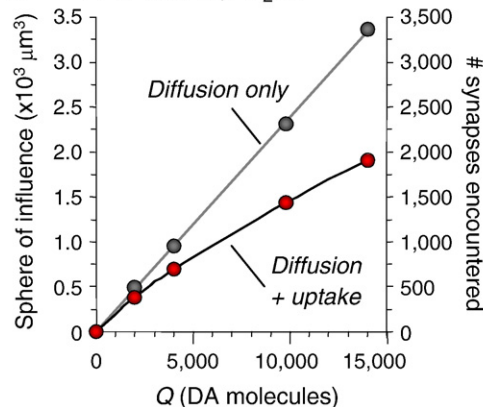
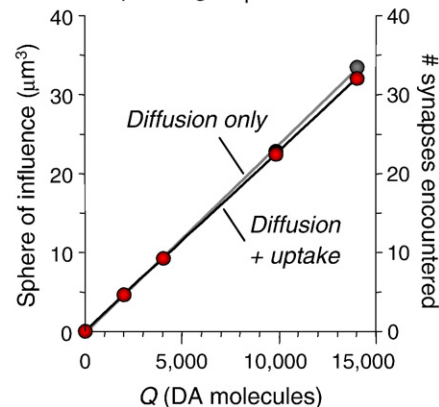
C ≥ 10 nM, e.g. D_2 RsD ≥ 1 μM, e.g. D_1 Rs

Fig. 4 – The sphere of influence of DA after quantal release in striatum and SNc depends on quantal size (Q) and DA receptor affinity. Sphere of DA influence, including number of synapses encountered vs. quantal size, Q , in (A,B) SNc and (C,D) striatum, with (diffusion + uptake) and without (diffusion only) region-specific DA uptake for activation of high-affinity (≥ 10 nM) and low-affinity (≥ 1 μM) DA receptors. A,B) In SNc, the sphere of influence of DA activation of either high-affinity (e.g., D_2 receptors, D_2 Rs), or low-affinity (e.g., D_1 receptors, D_1 Rs) was not affected by uptake, regardless of Q . C,D) In striatum, however, the sphere of influence for (C) high-affinity receptors was increasingly constrained by uptake as Q increased, consistent with a greater effect of the DAT on larger, longer-lasting DA transients. D) The sphere for low-affinity receptors was minimally affected by DAT-dependent uptake. In anatomical terms, the number of synapses contained in spheres defined by the $[DA]_o$ required to activate high-affinity receptors (A,C) in either SNc or striatum is 300–2500, whereas only 5–35 synapses would be encompassed within the sphere defined by the $[DA]_o$ required to activate low-affinity receptors (B,D). Panels C and D are modified from Cragg and Rice, 2004.

Our model shows that in SNc, the sphere of influence is directly proportional to the number of molecules released, whether specific DAT-mediated uptake is active or inhibited (Fig. 4A). For example, a seven-fold larger Q (14,000 vs. 2000) will have a seven-fold larger sphere of influence. A similar linear dependence on Q also occurs in striatum in the absence of uptake (Fig. 4C), although the resulting spheres are roughly 40% larger than those in SNc for equal Q , because of the smaller striatal volume fraction, α (compare Fig. 4A,B and C,D for “diffusion only”).

Normal DA uptake in striatum constrains the sphere of influence for activation of high-affinity DA receptors across a range of Q when compared to the spheres seen when clearance is by diffusion only (Fig. 4C). Importantly, the influence of

the DAT increases as Q increases, because larger, longer-lasting $[DA]_o$ transients allow longer times for uptake to remove DA (Cragg and Rice, 2004). For example, uptake decreases the sphere of influence by ~20% for $Q=2000$ DA molecules, but by ~40% for $Q=14,000$ (Fig. 4C). Strikingly, the spheres of influence for activation of low-affinity DA receptors are largely unaffected by uptake, regardless of Q , in both SNc (Fig. 4B) and striatum (Fig. 4D). Again, this reflects the rapid clearance of DA by diffusion and the consequently minimal influence of the DAT at $r < 2$ μm within which $[DA]_o$ is ≥ 1 μM.

The minimal effect of DA uptake on the spheres defined by activation of low-affinity DA receptors coupled with the smaller α of striatum vs. SNc means that the absolute tissue volume affected by a given Q is larger in striatum than in SNc.

In contrast, the spheres of influence for DA acting at high-affinity DA receptors in the presence of region-specific uptake are surprisingly similar between SNc and striatum after the release of equal Q (compare Fig. 4A and C for “diffusion+uptake”). This similarity reflects the opposing influences of uptake and α in each region: the larger α of SNc offsets the lower k' , whereas the higher k' of striatum is balanced by the smaller α . Intriguingly, the relative effects of these two parameters on sphere of influence vary with Q . For small, fast transients generated by the release of small Q , α dominates over uptake, such that for $Q=2000$ the sphere of influence is smaller in SNc than in striatum, whereas for larger transients larger Q , uptake dominates, such that for $Q=14,000$ the sphere is larger in SNc than in striatum.

4.1. How many synapses are within the sphere of DA influence in striatum and SNc?

Anatomical data suggest that there are on average 0.05 DA synapses per μm^3 or (1 DA synapse per $20 \mu\text{m}^3$) in the striatum (Doucet et al., 1986), although recent evaluation suggests that the density might be twice as high as that (Arbuthnott and Wickens, 2007). Even assuming a density of only 1 DA synapse per $20 \mu\text{m}^3$, with an average intersynaptic distance of $3.5 \mu\text{m}$, synaptic “spillover” to neighboring synapses within these spheres of influence readily occurs, with gradients of efficacy that depend on receptor sensitivity (Cragg and Rice, 2004). How many synapses will be affected? In striatum, the spheres of influence for $Q=2000$ – $14,000$ DA molecules acting at high-affinity receptors would contain ~20–100 DA synapses with normal DAT activity (Cragg and Rice, 2004). However, the sphere defined by $r \leq 2 \mu\text{m}$ for low-affinity DA receptors would affect only 0–1.5 DA synapses (Cragg and Rice, 2004).

Comparable calculations cannot be made for SNc, given the absence of axo-dendritic DA synapses and the presence of only few dendro-dendritic synapses in that region (Wilson et al., 1977). Moreover, in both SNc and striatum, DA receptors are expressed peri- and presynaptically at non-DA synapses, as well as on neuronal soma. A better sense of DA influence might therefore come from considering the number of receptors within a given sphere. However, such evaluation is hampered at present by a lack of quantitative data about receptor number for high and low-affinity DA receptors in striatum or SNc. A practical alternative approach is to consider the total population of synapses (DA and non-DA) potentially available for dopaminergic regulation. Total synaptic density in striatum is ~1 synapse per μm^3 (Pickel et al., 1981). Given the consistency of this synaptic density with that in other CNS regions, e.g., the CA1 region of hippocampus (Rusakov et al., 1999), we used this value to assess the influence of DA in SNc, as well as striatum. With normal uptake, released DA would encounter ~300 to 2500 synapses within the spheres defined by $[\text{DA}]_o \geq 10 \text{ nM}$ for $Q=2000$ – $14,000$ molecules in both regions (Fig. 4A,C). By contrast, the number of synapses in the spheres defined by $[\text{DA}]_o \geq 1 \mu\text{M}$ is nearly 200-fold lower, with only ~5 to 35 synapses encountered (Fig. 4B,D). Notably, the minimal influence of uptake on DA transients in these smaller spheres unmasks the effect of the larger α of SNc vs. striatum, with a lower number of synapses encountered in SNc than in striatum for all Q .

One final point to consider for both SNc and striatum is the active lifetimes of DA within these spheres of influence. Our model indicates that over the quantal range examined, the active lifetime during which $[\text{DA}]_o$ exceeds a minimum of 10 nM would be 10 to 100 ms. Notably, the duration of extracellular DA availability does not need to be shorter than this because the effector responses of DA receptors, which are G-protein coupled, range from 100 to 5000 ms (Cragg and Greenfield, 1997; Robinson and Caron, 1997; Gonon, 1997; Benoit-Marand et al., 2001; Phillips et al., 2002).

4.2. Temporal and spatial summation of DA quanta

In striatum, the greater effect of the DAT on larger, longer-lasting $[\text{DA}]_o$ transients after the release of larger Q (Fig. 4C), would also hold true for the release of multiple vesicles, from either single or multiple sites. The temporal and spatial summation of the resultant transients determine net $[\text{DA}]_o$ in the striatum and would therefore be expected to be strongly regulated by the DAT. Such a role is indeed consistent with the increases in basal $[\text{DA}]_o$ known to occur after uptake blockade (Cragg et al., 1997; Bradberry et al., 2000) or in DAT knockout mice (Giros et al., 1996). Importantly, the relative roles of the DAT on striatal $[\text{DA}]_o$ during tonic and phasic activity of midbrain DA neurons have also been determined experimentally (Floresco et al., 2003). As predicted by our model, inhibition of the DAT was found to cause a greater increase in $[\text{DA}]_o$ during enhanced burst firing of DA neurons, which leads to quantal summation, than during a simple increase in asynchronous tonic activity in these cells (Floresco et al., 2003).

5. Summary and implications for DA transmission in SNc and striatum

Our model of DA behavior after quantal release indicates that uptake does not limit initial DA overflow from a release site in either striatum or midbrain. This contrasts completely with regulation of the prototypical fast synaptic transmitter, glutamate (Rusakov and Kullmann, 1998; Barbour, 2001; Cragg and Rice, 2004). Moreover, diffusion, rather than uptake, is the key factor that regulates dynamic DA behavior, not only in the SNc, as predicted, but also in dorsal striatum in which regulation of $[\text{DA}]_o$ has been characterized as “uptake limited.” Nevertheless, this process constrains dopaminergic transmission in space and time: a DA transient produced by release of a single Q will have a maximum effective radius of 7–8 μm for activation of high-affinity DA receptors and an active lifetime of 10–100 ms. Thus, our model shows that diffusion and dilution, especially when aided by uptake, regulate dopaminergic transmission to a greater extent than sometimes imagined for signaling via volume transmission.

5.1. Volume transmission in the SNc

Evidence for DA volume transmission in the SNc (Fig. 5) includes the predominance of extrasynaptic DA receptors (Cameron and Williams, 1993; Sesack et al., 1994; Yung et al., 1995), the virtual absence of DA synapses (Wilson et al., 1977), and a limited influence of DA uptake on either exogenously

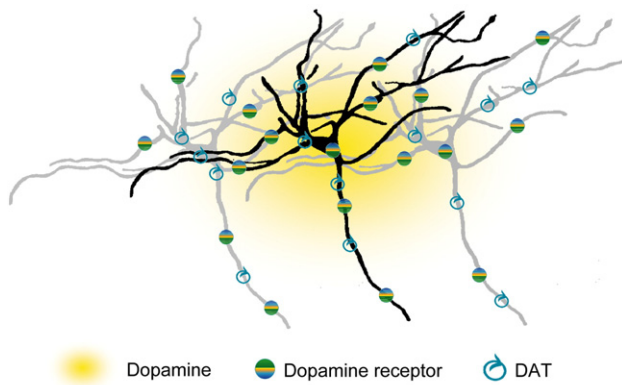


Fig. 5 – Volume transmission of DA in the SNc. Schematic representation of the influence of somatodendritically released DA from one DA neuron (black) diffusing to activate DA receptors (D_2 autoreceptors) in the releasing cell, as well as on the soma and dendrites of neighboring DA neurons (gray). The representations of DA receptors and DATs in this diagram do not reflect actual densities; the DA neuron is modified from a drawing by Fallon et al., 1978.

diffusing (Cragg et al., 2001) or endogenously released DA (Cragg et al., 1997; Chen and Rice, 2001). Moreover, as discussed here and elsewhere (Rice, 2000; Cragg et al., 2001), the release of a single vesicle of DA could modulate the excitability of tens to thousands of synapses within distances a few micrometers of a release site, in both SNc and striatum. This broad influence of released DA meets the definition of volume transmission given in the Introduction, which is independent of the source of release, i.e., from synaptic or non-synaptic sites.

Recently, the view that DA acts by volume transmission in the SNc has been challenged, based on electrophysiological monitoring of D_2 receptor-mediated inhibitory “postsynaptic” currents (IPSCs) in DA neurons (Beckstead et al., 2004, 2007). In these studies, local electrical stimulation in the presence of a cocktail of glutamate and GABA receptor antagonists evoked a robust IPSC in most DA neurons examined. Beckstead, Williams and colleagues have argued against DA volume transmission in the SNc (Beckstead et al., 2004, 2007) from their observation that the time course of DA-dependent IPSCs is unaltered under conditions that increased the amplitude of the IPSCs, including by increasing stimulation intensity (Figs. 6A,B) or increasing quantal content with L-DOPA. Their interpretation is based on the premise that, “If super-threshold concentrations of dopamine were typically traveling micrometers to reach target D_2 receptors then increasing release would have prolonged the time course of the synaptic current” (Beckstead et al., 2007). There are two flaws in this premise, however. First, there are few dendro-dendritic synapses in the SN (Wilson et al., 1977) that could mediate synaptic DA transmission; moreover, no synaptic D_2 receptors have been identified in midbrain DA neurons (Sesack et al., 1994; Yung et al., 1995; Khan et al., 1998). Second, the authors’ premise does not take into account the time course of diffusion of DA from a release site to its site of action. As shown in this review, the time to peak and duration of $[DA]_o$ transients after quantal release are much shorter than those of recorded IPSCs. At a

distance of 5 μm from a release site, the resulting $[DA]_o$ transient would reach a maximum ~ 10 ms after release, whereas the observed peak IPSC does not occur until ~ 500 ms after the stimulus (Fig. 6B; $[DA]_o$ data from Fig. 1 superimposed on IPSC records from Beckstead et al., 2004). Even 20 μm from a release site, peak $[DA]_o$ would occur only 250 ms after release, which is still half the time of the IPSC peak (Fig. 6B). Thus, the IPSC data lack the temporal resolution to provide insight into the role of volume transmission on DA communication on the SN.

The rate-limiting step in IPSC generation is activation of D_2 receptor-linked GIRK (G-protein coupled inward rectifier K^+) channels, not DA diffusion (Beckstead et al., 2004). The IPSC

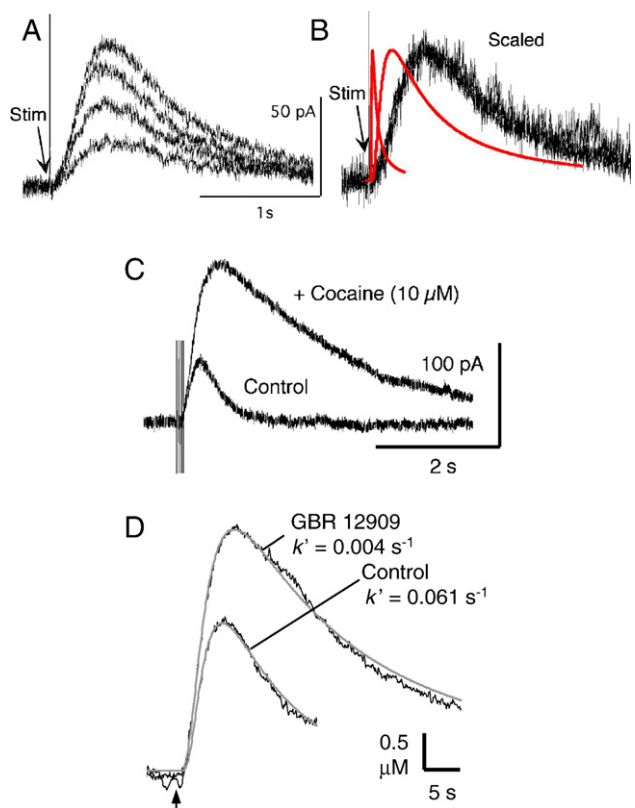


Fig. 6 – Comparison of the time course of IPSCs and DA diffusion in the SNc. A) Increasing current amplitude of D_2 autoreceptor-mediated IPSCs evoked in a midbrain DA neuron by local stimulation with increasing stimulus intensity (modified from Beckstead et al., 2004). B) The time course of the IPSCs is unaffected by increasing current amplitude, as seen when the current records in (A) were normalized to their maxima (black lines) (modified from Beckstead et al., 2004). Superimposed on these IPSCs are normalized DA diffusion curves (red lines) modeled for $r=5$ and $r=20$ μm from a release site in SNc (from Fig. 1). This comparison shows that the time course of DA transients is much faster than receptor-mediated signaling. C) Increased time course and amplitude of evoked IPSCs in a midbrain DA neuron when the DAT is inhibited by cocaine (modified from Beckstead et al., 2004). D) Increased time course and amplitude of experimentally monitored DA diffusion curves in the SNc when the DAT is inhibited by GBR 12909 ($r=100$ μm) (modified from Cragg et al., 2001).

data reported are consistent with previous studies of the time course of D_2 autoreceptor-mediated inhibition of DA release in striatum, in which autoinhibition begins 50–100 ms after DA single-pulse evoked DA release, then persists for up to 5 s afterwards, which greatly exceeds the lifetime of the DA transient that initiated the autoreceptor response (Phillips et al., 2002). It is unsurprising, therefore, that the time course of DA IPSCs would be independent of the amplitude of rapid $[DA]_o$ transients that initiate D_2 receptor-linked GIRK channel opening. Indeed, the dependence of IPSC amplitude on stimulation intensity supports a role for DA volume transmission, rather than arguing against it. Local stimulation activates not just the recorded neuron, but also surrounding cells and dendrites, such that the resulting increase in $[DA]_o$ is a population response. As stimulus intensity is increased, the resulting increase in current spread will lead to action potential generation and somatodendritic DA release from increasingly distant neurons. The dependence of D_2 autoreceptor-mediated current amplitude on stimulus intensity (Fig. 6A) is therefore consistent with increases in $[DA]_o$ detected by the recorded cell as the summation of DA transients from both local and increasingly distant sources, reflecting volume transmission.

Given the likely contribution of distant release sites, and therefore multiple vesicles, to somatodendritic $[DA]_o$ transients, it is also unsurprising that one condition that did increase IPSC time course in DA neurons was DAT inhibition by cocaine (Beckstead et al., 2004, 2007) (Fig. 6C). This response is consistent with previous studies of the effect of DAT inhibition on experimental DA diffusion profiles in the SNc recorded 100 μ m from a pressure ejection pipette (Cragg et al., 2001) (Fig. 6D). The proportionately greater effect of DAT inhibition on the

IPSC records than on DA diffusion curves reflects the primary contribution of uptake to DA clearance after release by electrical stimulation, in which a locally homogeneous increase in $[DA]_o$ is predominantly cleared by uptake rather than diffusion (Garris et al., 1994). As Beckstead et al. (2004) conclude, their effect of cocaine confirms “the crucial role of uptake transporters in limiting the time dopamine is in the extracellular space,” and therefore available to mediate volume transmission.

5.2. A new model of the striatal DA synapse

The conventional representation of a striatal DA synapse shows vesicular release with re-uptake of release DA into the presynaptic terminal by perisynaptic DATs, which restricts synaptic spillover of DA and regulates synaptic DA concentration (Fig. 7, left panel). However, our kinetic analysis of DA diffusion and uptake after quantal release shows that synaptic DA spillover cannot be gated by perisynaptic DATs, because the kinetics of DA transport cannot compete with the rapid escape of DA away from a release site by diffusion (Cragg and Rice, 2004). This further means that perisynaptic DATs can do little to alter the concentration of DA within a striatal synapse. It is therefore time for a significant revision of the depiction of a DA synapse that incorporates these inconvenient facts (Fig. 7, right panel).

Two main factors contribute to the dominance of diffusion over DAT-mediated uptake. The first is the rapidity of diffusion over micrometer distances and the low DAT cycle rate (2–5 DA molecules/s per transporter at 37 °C) (Povlock and Schenk, 1997; Prasad and Amara, 2001). The second, less obvious factor is that DAT expression is limited to DA neurons,

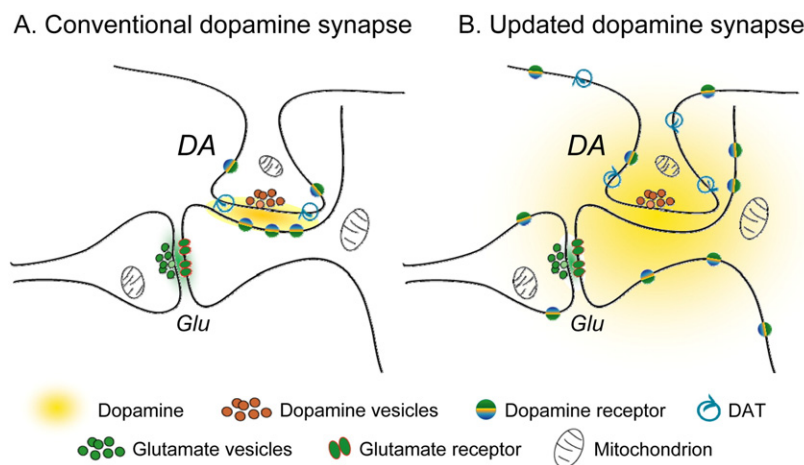


Fig. 7 – Volume transmission of DA in the striatum. A) Schematic representation of a conventional DA synapse on the neck of a dendritic spine on a striatal medium spiny neuron. Glutamatergic input forms synapses on the heads of these spines. Synaptically released DA is often depicted as being constrained within a synapse by perisynaptic DATs, with primary activation of subsynaptic DA receptors, like the regulation of glutamate (Glu) at a glutamatergic synapse. The extrasynaptic localization of DATs and DA receptors, as well as the rapid time course of DA diffusion away from a synapse compared to DAT-mediated uptake makes this “conventional” model synapse inaccurate and outdated. B) Schematic representation of an updated DA synapse that shows ready DA spillover, unconstrained by DATs. The cloud of released DA, diffusing in three dimensions, encounters predominantly extrasynaptic DA autoreceptors on DA axons and heteroreceptors on neighboring cells, including medium spiny neurons. Released DA will be taken up only when it encounters DATs on DA axons. The representations of DA receptors and DATs in A and B do not reflect actual densities.

including DA axons in striatum (Nirenberg et al., 1996). Immediately after release, therefore, a DA molecule diffusing in three-dimensional space will be subject to transporter activity only if it encounters the presynaptic membrane of the releasing DA axon (Fig. 7, right panel). This geometric organization is analogous to that of the retina, in which DA is released from DAT-expressing amacrine cells oriented in a single plane in the inner plexiform layer: once a DA molecule escapes this plane, its behavior is governed only by diffusion (Witkovsky et al., 1993; Rice, 2000). Diffusing DA can readily interact with predominantly extrasynaptic DA receptors (Sesack et al., 1994; Yung et al., 1995; Hersch et al., 1995; Khan et al., 1998) (Fig. 7, right panel), until it reaches the boundary set by receptor sensitivities (e.g., Fig. 2).

The perisynaptic landscape of a glutamate synapse is very different from that for DA synapses (Cragg and Rice, 2004). Glutamate transporters are localized primarily on non-glutamatergic cells, including astrocytic processes that can ensheath a glutamate synapse, as well as postsynaptic neurons (Seal and Amara, 1999; Danbolt, 2001). Moreover, glutamate transport kinetics exceed those of the DAT by an order of magnitude, with transport-cycle rates of over 35 glutamate molecules/s per transporter at 37 °C (EAAT2) (Wadiche et al., 1995). Most glutamatergic communication is mediated by fast-responding (1–2 ms), ionotropic glutamate receptors located immediately postsynaptically, with slower modulation via metabotropic receptors found near the perimeter of the synapse (Ottersen and Landsend, 1997) and possible additional contributions from extrasynaptic receptors (Galvan et al., 2006). Although there is experimental evidence to support glutamate spillover (Rusakov and Kullmann, 1998; Rusakov et al., 1999), released glutamate is typically constrained by uptake to remain perisynaptic, thus preserving synaptic independence (e.g., Barbour (2001)).

At the present time, DA synapses are most commonly illustrated as though they were glutamate synapses (Fig. 7, left panel), despite marked differences between the structural and functional anatomy of these transmitter release sites. To understand dopaminergic transmission requires that the classic representation of the DA synapse be updated to reflect the functional microanatomy of these synapses, which, unlike glutamate synapses, are designed for overflow and volume transmission (Fig. 7, right panel).

5.3. Conclusions

The slow kinetics and limited distribution of the DAT leads to predominant regulation of DA behavior by diffusion in both SNc and striatum. Following quantal release, nigrostriatal DA can readily spill out from release sites and act within effective radii of several micrometers. Thus, diffusion-based volume transmission is the predominant mode of DA signaling after both somatodendritic and synaptic release.

Acknowledgments

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REFERENCES

- Arbuthnott, G.W., Wickens, J., 2007. Space, time and dopamine. *Trends Neurosci.* 30, 62–69.
- Barbour, B., 2001. An evaluation of synapse independence. *J. Neurosci.* 21, 7969–7984.
- Beckstead, M.J., Grandy, D.K., Wickman, K., Williams, J.T., 2004. Vesicular dopamine release elicits an inhibitory postsynaptic current in midbrain dopamine neurons. *Neuron* 42, 939–946.
- Beckstead, M.J., Ford, C.P., Phillips, P.E., Williams, J.T., 2007. Presynaptic regulation of dendrodendritic dopamine transmission. *Eur. J. Neurosci.* 26, 1479–1488.
- Benoit-Marand, M., Borrelli, E., Gonon, F., 2001. Inhibition of dopamine release via presynaptic D2 receptors: time course and functional characteristics in vivo. *J. Neurosci.* 21, 9134–9141.
- Bergquist, F., Shahabi, H.N., Nissbrandt, H., 2003. Somatodendritic dopamine release in rat substantia nigra influences motor performance on the accelerating rod. *Brain Res.* 973, 81–91.
- Bradberry, C.W., Barrett-Larimore, R.L., Jatlow, P., Rubino, S.R., 2000. Impact of self-administered cocaine and cocaine cues on extracellular dopamine in mesolimbic and sensorimotor striatum in rhesus monkeys. *J. Neurosci.* 20, 3874–3883.
- Cameron, D.L., Williams, J.T., 1993. Dopamine D1 receptors facilitate transmitter release. *Nature* 366, 344–347.
- Chen, B.T., Rice, M.E., 2001. Novel Ca²⁺ dependence and time course of somatodendritic dopamine release: substantia nigra versus striatum. *J. Neurosci.* 21, 7841–7847.
- Ciliax, B.J., Heilman, C., Demchyshyn, L.L., Pristupa, Z.B., Ince, E., Hersch, S., Niznik, H.B., Levey, A.I., 1995. The dopamine transporter: immunocytochemical characterization and localization in brain. *J. Neurosci.* 15, 1714–1723.
- Colliver, T.L., Pyott, S.J., Achalabun, M., Ewing, A.G., 2000. VMAT-mediated changes in quantal size and vesicular volume. *J. Neurosci.* 20, 5276–5282.
- Cragg, S.J., Greenfield, S.A., 1997. Differential autoreceptor control of somatodendritic and axon terminal dopamine release in substantia nigra, ventral tegmental area, and striatum. *J. Neurosci.* 17, 5738–5746.
- Cragg, S.J., Rice, M.E., 2004. Dancing past the DAT at a DA synapse. *Trends Neurosci.* 27, 270–277.
- Cragg, S.J., Rice, M.E., Greenfield, S.A., 1997. Heterogeneity of electrically evoked dopamine release and reuptake in substantia nigra, ventral tegmental area, and striatum. *J. Neurophysiol.* 77, 863–873.
- Cragg, S.J., Nicholson, C., Kume-Kick, J., Tao, L., Rice, M.E., 2001. Dopamine-mediated volume transmission in midbrain is regulated by distinct extracellular geometry and uptake. *J. Neurophysiol.* 85, 1761–1771.
- Danbolt, N.C., 2001. Glutamate uptake. *Prog. Neurobiol.* 65, 1–105.
- Descarries, L., Watkins, K.C., Garcia, S., Bosler, O., Doucet, G., 1996. Dual character, asynaptic and synaptic, of the dopamine innervation in adult rat neostriatum: a quantitative autoradiographic and immunocytochemical analysis. *J. Comp. Neurol.* 375, 167–186.
- Doucet, G., Descarries, L., Garcia, S., 1986. Quantification of the dopamine innervation in adult rat neostriatum. *Neuroscience* 19, 427–445.
- Fallon, J.H., Riley, J.N., Moore, R.Y., 1978. Substantia nigra dopamine neurons: separate populations project to neostriatum and allocortex. *Neurosci. Lett.* 7, 157–162.
- Falkenburger, B.H., Barstow, K.L., Mintz, I.M., 2001. Dendrodendritic inhibition through reversal of dopamine transport. *Science* 293, 2465–2470.
- Floresco, S.B., West, A.R., Ash, B., Moore, H., Grace, A.A., 2003. Afferent modulation of dopamine neuron firing differentially regulates tonic and phasic dopamine transmission. *Nat. Neurosci.* 6, 968–973.

- Fuxe, K., Agnati, L.F., 1991. *Volume Transmission in the Brain*. Raven Press, New York.
- Galvan, A., Kuwajima, M., Smith, Y., 2006. Glutamate and GABA receptors and transporters in the basal ganglia: what does their subsynaptic localization reveal about their function? *Neuroscience* 143, 351–375.
- Garris, P.A., Ciolkowski, E.L., Pastore, P., Wightman, R.M., 1994. Efflux of dopamine from the synaptic cleft in the nucleus accumbens of the rat brain. *J. Neurosci.* 14, 6084–6093.
- Giros, B., Jaber, M., Jones, S.R., Wightman, R.M., Caron, M.G., 1996. Hyperlocomotion and indifference to cocaine and amphetamine in mice lacking the dopamine transporter. *Nature* 379, 606–612.
- Gonon, F., 1997. Prolonged and extrasynaptic excitatory action of dopamine mediated by D1 receptors in the rat striatum in vivo. *J. Neurosci.* 17, 5972–5978.
- Groves, P.M., Linder, J.C., Young, S.J., 1994. 5-Hydroxydopamine-labeled dopaminergic axons: three-dimensional reconstructions of axons, synapses and postsynaptic targets in rat neostriatum. *Neuroscience* 58, 593–604.
- Hersch, S.M., Ciliax, B.J., Gutekunst, C.A., Rees, H.D., Heilman, C.J., Yung, K.K., Bolam, J.P., Ince, E., Yi, H., Levey, A.I., 1995. Electron microscopic analysis of D1 and D2 receptor proteins in the dorsal striatum and their synaptic relationships with motor corticostriatal afferents. *J. Neurosci.* 15, 5222–5237.
- Hersch, S.M., Yi, H., Heilman, C.J., Edwards, R.H., Levey, A.I., 1997. Subcellular localization and molecular topology of the dopamine transporter in the striatum and substantia nigra. *J. Comp. Neurol.* 388, 211–227.
- Jaffe, E.H., Marty, A., Schulte, A., Chow, R.H., 1998. Extrasynaptic vesicular transmitter release from the somata of substantia nigra neurons in rat midbrain slices. *J. Neurosci.* 18, 3548–3553.
- Khan, Z.U., Mrzljak, L., Gutierrez, A., de la Calle, A., Goldman-Rakic, P.S., 1998. Prominence of the dopamine D₂ short isoform in dopaminergic pathways. *Proc. Natl. Acad. Sci. U. S. A.* 95, 7731–7736.
- Neve, K.A., Neve, R.L., 1997. Molecular biology of dopamine receptors. In: Neve, K.A., Neve, R.L. (Eds.), *The Dopamine Receptors*. Humana Press, Totowa, NJ, pp. 27–76.
- Nicholson, C., 1985. Diffusion from an injected volume of a substance in brain tissue with arbitrary volume fraction and tortuosity. *Brain Res.* 333, 325–329.
- Nicholson, C., 1995. Interaction between diffusion and Michaelis–Menten uptake of dopamine after iontophoresis in striatum. *Biophys. J.* 68, 1699–1715.
- Nicholson, C., Syková, E., 1998. Extracellular space structure revealed by diffusion analysis. *Trends Neurosci.* 21, 207–215.
- Nirenberg, M.J., Vaughan, R.A., Uhl, G.R., Kuhar, M.J., Pickel, V.M., 1996. The dopamine transporter is localized to dendritic and axonal plasma membranes of nigrostriatal dopaminergic neurons. *J. Neurosci.* 16, 436–447.
- Nirenberg, M.J., Chan, J., Vaughan, R.A., Uhl, G.R., Kuhar, M.J., Pickel, V.M., 1997. Immunogold localization of the dopamine transporter: an ultrastructural study of the rat ventral tegmental area. *J. Neurosci.* 17, 5255–5262.
- Ottersen, O.P., Landsend, A.S., 1997. Organization of glutamate receptors at the synapse. *Eur. J. Neurosci.* 9, 2219–2224.
- Phillips, P.E.M., Hancock, P.J., Stamford, J.A., 2002. Time window of autoreceptor-mediated inhibition of limbic and striatal dopamine release. *Synapse* 44, 15–22.
- Pickel, V.M., Beckley, S.C., Joh, T.H., Reis, D.J., 1981. Ultrastructural immunocytochemical localization of tyrosine hydroxylase in the neostriatum. *Brain. Res.* 225, 373–385.
- Pothos, E.N., Davila, V., Sulzer, D., 1998. Presynaptic recording of quanta from midbrain dopamine neurons and modulation of the quantal size. *J. Neurosci.* 18, 4106–4118.
- Povlock, S.L., Schenk, J.O., 1997. A multisubstrate kinetic mechanism of dopamine transport in the nucleus accumbens and its inhibition by cocaine. *J. Neurochem.* 69, 1093–1105.
- Prasad, B.M., Amara, S.G., 2001. The dopamine transporter in mesencephalic cultures is refractory to physiological changes in membrane voltage. *J. Neurosci.* 21, 7561–7567.
- Rice, M.E., 2000. Distinct regional differences in dopamine-mediated volume transmission. In: Agnati, L.F., Fuxe, K., Nicholson, C., Sykova, E. (Eds.), *Volume Transmission Revisited*, vol. 125. Elsevier Science, pp. 275–288.
- Rice, M.E., Nicholson, C., 1991. Diffusion characteristics and extracellular volume fraction during normoxia and hypoxia in slices of rat neostriatum. *J. Neurophysiol.* 65, 264–272.
- Richfield, E.K., Penney, J.B., Young, A.B., 1989. Anatomical and affinity state comparisons between dopamine D1 and D2 receptors in the rat central nervous system. *Neuroscience* 30, 767–777.
- Robertson, G.S., Robertson, H.A., 1989. Evidence that L-dopa-induced rotational behavior is dependent on both striatal and nigral mechanisms. *J. Neurosci.* 9, 3326–3331.
- Robinson, S.W., Caron, M.G., 1997. Selective inhibition of adenylyl cyclase type V by the dopamine D3 receptor. *Mol. Pharmacol.* 52, 508–514.
- Ross, S.B., 1991. Synaptic concentration of dopamine in the mouse striatum in relationship to the kinetic properties of the dopamine receptors and uptake mechanism. *J. Neurochem.* 56, 22–29.
- Rusakov, D.A., Kullmann, D.M., 1998. Extrasynaptic glutamate diffusion in the hippocampus: ultrastructural constraints, uptake, and receptor activation. *J. Neurosci.* 18, 3158–3170.
- Rusakov, D.A., Kullmann, D.M., Stewart, M.G., 1999. Hippocampal synapses: do they talk to their neighbours? *Trends Neurosci.* 22, 382–388.
- Seal, R.P., Amara, S.G., 1999. Excitatory amino acid transporters: a family in flux. *Annu. Rev. Pharmacol. Toxicol.* 39, 431–456.
- Sesack, S.R., Aoki, C., Pickel, V.M., 1994. Ultrastructural localization of D2 receptor-like immunoreactivity in midbrain dopamine neurons and their striatal targets. *J. Neurosci.* 14, 88–106.
- Stamford, J.A., Kruk, Z.L., Palij, P., Millar, J., 1988. Diffusion and uptake of dopamine in rat caudate and nucleus accumbens compared using fast cyclic voltammetry. *Brain Res.* 448, 381–385.
- Sulzer, D., Pothos, E.N., 2000. Regulation of quantal size by presynaptic mechanisms. *Rev. Neurosci.* 11, 159–212.
- Timmerman, W., Abercrombie, E.D., 1996. Amphetamine-induced release of dendritic dopamine in substantia nigra pars reticulata: D1-mediated behavioral and electrophysiological effects. *Synapse* 23, 280–291.
- Wadiche, J.I., Ariza, J.L., Amara, S.G., Kavanaugh, M.P., 1995. Kinetics of a human glutamate transporter. *Neuron* 14, 1019–1027.
- Wilson, C.J., Groves, P.M., Fikova, E., 1977. Monoaminergic synapses, including dendro-dendritic synapses in the rat substantia nigra. *Exp. Brain Res.* 30, 161–174.
- Witkovsky, P., Nicholson, C., Rice, M., Bohmaker, K., Meller, E., 1993. Extracellular dopamine concentration in the retina of the clawed frog, *Xenopus laevis*. *Proc. Natl. Acad. Sci. USA* 90, 5667–5671.
- Yung, K.K.L., Bolam, J.P., Smith, A.D., Hersch, S.M., Ciliax, B.J., Levey, A.I., 1995. Immunocytochemical localization of D1 and D2 dopamine receptors in the basal ganglia of the rat: light and electron microscopy. *Neuroscience* 65, 709–730.